

E.1.1 Di et al. (2017)

Di et al. (2017) evaluated the relationship between long-term PM_{2.5} exposure and all-cause mortality in nearly 61 million U.S. Medicare enrollees (over the age of 64) through 460 million person-years of follow-up and roughly 22 million observed deaths. This cohort comprised approximately 15% of the total U.S. population, included people living in rural areas, and is one of the largest cohort studies published to date. The authors modeled PM_{2.5} exposure across the contiguous U.S. using a hybrid methodology that included land use regression, satellite data, and monitor data, and resolved estimations to 1 x 1-kilometer areas. Di et al. (2017) used Cox proportional-hazards models with a generalized estimating equation. Adjustment for potential confounding by the co-pollutant O₃ was performed, which slightly attenuated the relationship between PM_{2.5} and mortality. The authors also performed statistical testing of the potential for non-linear effects and concluded that the data supported a nearly-linear concentration-response relationship with no signal of a threshold down to at least 5 µg/m³.

All-Cause Mortality

In a two-pollutant model, the coefficient and standard error for PM_{2.5} are estimated from the hazard ratio (1.073) and 95% confidence interval of (1.071-1.075) associated with a change in annual mean PM_{2.5} exposure of 10.0 µg/m³ (Di et al., 2017, Table 2 Main Analysis, Cox PH with GEE).

E.1.2 Turner et al. (2016)

Turner et al. (2016) examined the relationship between long-term PM_{2.5} exposure (1982-2004) and mortality (all-cause, cause-specific) in American Cancer Society Cancer Prevention Study-II participants (aged 30-99 years). Estimated PM_{2.5} concentrations were obtained using monthly PM_{2.5} monitor data (1999-2008) and a national-level hybrid land use regression (LUR) and Bayesian maximum entropy (BME) interpolation model. Turner et al. (2016) utilized random-effects Cox proportional hazard models adjusted *a priori* for individual, population subgroups, and ecological variables. In addition to adjusting for individual-level and ecological covariates, Turner et al. (2016) also controlled for occupational PM_{2.5} exposure and adjusted for the potential co-pollutants O₃ and nitrogen dioxide.

All-Cause Mortality

In a multi-pollutant model, the coefficient and standard error for PM_{2.5} are estimated from the hazard ratio (1.06) and 95% confidence interval of (1.04–1.08) associated with a change of 10.0 µg/m³ in the mean PM_{2.5} exposure level from 1999-2004 (Turner et al., 2016, Table E10 HBM PM_{2.5}, MP model, controlling for HBM O₃ 1982-2004).

E.1.3 Woodruff et al. (2008)

Woodruff et al. (2008) examined the relationship between long-term exposure to fine PM_{2.5} air pollution and post-neonatal infant mortality in 3,583,495 births from 96

counties containing >249,999 residents across the U.S. between 1999-2002 using data from the National Center for Health Statistics (NCHS). They linked average PM_{2.5} monitoring data over the first two months of life with 6,639 post neonatal deaths, using logistic regression that incorporated generalized estimating equations (GEE) to estimate the odds ratios for all-cause and cause-specific post-neonatal mortality by exposure to air pollution. The study population experienced a median PM_{2.5} concentration of 14.8 µg/m³, with 25% of the population experiencing concentrations below 12 µg/m³ and above 18.8 µg/m³. The study included an evaluation of the appropriateness of a linear form from analysis based on quartiles of exposure and identified the linear form as a reasonable assumption.

All-Cause Mortality

In a single-pollutant model, the coefficient and standard error for PM_{2.5} are estimated from the odds ratio (1.04) and 95% confidence interval of (0.98–1.11) associated with a change of 7 µg/m³ in the mean PM_{2.5} exposure level during the first two months of life (Woodruff et al., 2008, Table 4 PM_{2.5} single-pollutant model, all causes).

E.1.4 Pope et al. (2019)

Pope et al. (2019) examined the relationship between long-term PM_{2.5} exposure and all-cause mortality in a cohort of 1,599,329 U.S. adults (aged 18-84 years) who were interviewed in the National Health Interview Surveys (NHIS) between 1986 and 2014 and linked to the National Death Index (NDI) through 2015. The authors also constructed a subcohort of 635,539 adults from the full cohort for whom body mass index (BMI) and smoking status data were available. They employed a hybrid modeling technique to estimate annual-average PM_{2.5} concentrations derived from regulatory monitoring data and constructed in a universal kriging framework using geographic variables including land use, population, and satellite estimates. Pope et al. (2019) assigned annual-average PM_{2.5} exposure from 1999-2015 to each individual by census tract and utilized complex (accounting for NHIS's sample design) and simple Cox proportional hazards models for the full cohort and the subcohort. We report the results of the complex model for the subcohort, which controls for individual-level covariates including age, sex, race-ethnicity, inflation-adjusted income, education level, marital status, rural versus urban, region, survey year, BMI, and smoking status.

All-Cause Mortality

In a single-pollutant model, the coefficient and standard error for PM_{2.5} are estimated from the hazard ratio (1.12) and 95% confidence interval (1.08-1.15) associated with a change in annual mean PM_{2.5} exposure of 10.0 µg/m³ (Pope et al. 2019, Table 2, Subcohort).

E.1.5 Wu et al. (2020)

Wu et al. (2020) evaluated the relationship between long-term PM_{2.5} exposure and all-cause mortality in more than 68.5 million Medicare enrollees (over the age of 64), using Medicare claims data from 2000-2016 representing over 573 million person-years of

follow-up and over 27 million deaths. This cohort included over 20% of the U.S. population and was, at the time of publishing, the largest air pollution study cohort to date. The authors modeled PM_{2.5} exposure at a 1-km² grid resolution using a hybrid ensemble-based prediction model that combined three machine learning models and relied on satellite data, land-use information, weather variables, chemical transport model simulation outputs, and monitor data. Wu et al. (2020) fit five different statistical models: a Cox proportional hazards model, a Poisson regression model, and three causal inference approaches (GPS estimation, GPS matching, and GPS weighting). All five statistical approaches provided consistent results; we report the results of the Cox proportional hazards model here. The authors adjusted for numerous individual-level and community-level confounders, and sensitivity analyses suggest that the results are robust to unmeasured confounding bias.

All-Cause Mortality

In a single-pollutant model, the coefficient and standard error for PM_{2.5} are estimated from the hazard ratio (1.066) and 95% confidence interval (1.058-1.074) associated with a change in annual mean PM_{2.5} exposure of 10.0 µg/m³ (Wu et al. 2020, Table S3, Main analysis, 2000-2016 Cohort, Cox PH).

E.1.6 Chen et al. (2023)

Chen et al. (2023) examined the relationship between long-term exposure to low-level PM_{2.5} and mortality across three studies that evaluated impacts on eight different cohorts based in either North America or Europe. The study harmonizes the analysis across the three major studies for factors such as exposure, confounders, and outcome definition to assess the effect of differences in study design on mortality impact. The authors used meta-analysis to estimate a combined concentration-response function (CRF) based on the extended shape-constrained health impact function (eSCHIF). The combined design included over 81 million participants and 692 million person-years of follow-up across the eight cohorts. Differences across the studies were reduced slightly by adjusting for a more complete set of confounding variables and harmonizing exposure. The meta-analysis-based estimate resulted in a monotonically increasing function showing effects down to the lowest observed levels of PM_{2.5} (3.7 µg/m³). The curve was substantially influenced by the U.S. Medicaid cohort at lower PM_{2.5} concentrations, and by the Swiss cohort at higher levels.

All-Cause Mortality

Because the effects of harmonization were minor, result of the meta-analysis using the original reported values was employed to derive the health impact function. In a single-pollutant model, the coefficient and standard error for PM_{2.5} are estimated from the odds ratio (1.05) and 95% confidence interval of (1.03–1.08) associated with a change of 5 µg/m³ in the mean annual PM_{2.5} exposure level (Chen et al., 2023, Figure S1, Random-effects meta-analysis of eight cohorts, Original Report value).

E.2 Chronic/Severe Illness

Table E-2 below summarizes the health impact functions used to estimate the relationship between PM_{2.5} and chronic / severe health effects. We present a brief summary of each of the studies below.

Table E-2. Core Health Impact Functions for Particulate Matter and Chronic Illness

Effect	Author	Year	Location	Age	Co-Poll	Metric	Beta	Std Err	Form	Notes
Cardiac Arrest	Ensor et al.	2013	Houston, TX	18-99		D24HourMean	0.006376	0.002823	Logistic	
Cardiac Arrest	Rosenthal et al.	2008	Indianapolis, Indiana	0-99		D24HourMean	0.00198	0.005018	Logistic	
Cardiac Arrest	Silverman et al.	2010	New York City	0-99		D24HourMean	0.003922	0.00222	Logistic	
Lung Cancer	Gharibvand et al.	2017	Nationwide U.S. and 5 Canadian provinces	30-99		Mean	0.037844	0.013121	Log-linear	
Alzheimer's Disease	Kioumourtzoglou et al	2016	50 Northeastern U.S. cities	65-99		Mean	0.139762	0.017753	Log-linear	
Parkinson's Disease	Kioumourtzoglou et al	2016	50 Northeastern U.S. cities	65-99		Mean	0.076961	0.018905	Log-linear	
Stroke	Kloog et al.	2012	New England	65-99		Mean	0.00343	0.001265	Log-linear	
Acute Myocardial Infarction, Nonfatal	Peters et al.	2001	Boston, MA	18-99		D24HourMean	0.024121	0.009285	Logistic	
Acute Myocardial Infarction, Nonfatal	Pope et al.	2006	Greater Salt Lake City, UT	18-99		D24HourMean	0.0048	0.0019	Logistic	Index MI and unstable angina
Acute Myocardial Infarction, Nonfatal	Sullivan et al.	2005	King County, WA	18-99		D24HourMean	0.0019	0.0022	Logistic	
Acute Myocardial Infarction, Nonfatal	Wei et al.	2019	Continental U.S.	65-99		D24HourMean	0.0011	0.0002	Logistic	
Acute Myocardial Infarction, Nonfatal	Zanobetti and Schwartz	2006	Greater Boston, MA	18-99		D24HourMean	0.0053	0.0022	Logistic	Age range adjusted Admissions through ER visits only.
Acute Myocardial Infarction, Nonfatal	Zanobetti et al.	2009	26 U.S. Comm	18-99		D24HourMean	0.0022	0.0006	Log-linear	Age range adjusted. All Seasons.